2024 PSYC489 Research Project

To Fathom Hell or Soar Angelic: Reflexive Thematic Analysis of Self-Reported Adverse Psilocybin and MDMA Experience in Women

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Introduction

Amid widespread mental health challenges, insufficient resources and inadequate treatment options, the therapeutic potential of psychedelics[1] has recently garnered renewed clinical and public interest. As of February 2024, over one thousand clinical trials involving psychedelic substances were registered at various stages of development (World Health Organisation [WHO], 2024). This resurgence is primarily driven by promising clinical-based research of rapid onset, sustained therapeutic effects with minimal dosing, and lower risks of side effects compared to traditional treatments (Andersen et al., 2021; Gill et al., 2020; Mitchell & Anderson, 2024). Two substances in particular, psilocybin and MDMA[2], are at the forefront of this movement prompting global policy reforms to reclassify them as viable clinical interventions (Heifets & Malenka, 2019; Holze et al., 2024). The United States Food and Drug Administration (FDA) granted breakthrough therapy designations to MDMA in 2017 and psilocybin in 2019 (Heifets & Malenka, 2019), and a significant milestone was reached in July 2023 when Australia legalized the psychiatric prescription of MDMA for post-traumatic stress disorder (PTSD) and psilocybin for treatment-resistant depression (TRD) (WHO, 2024).

Despite these promising developments, in August 2024, the FDA rejected MDMA as a psychiatric intervention for PTSD, citing concerns about clinical trial design and requesting additional large-scale studies (Stringer, 2024). This decision reflects apprehensions that clinical data may be overestimating the therapeutic benefits of these substances while underreporting or omitting rare but significant adverse effects. Concerns stem from a variety of factors including small and/or heterogeneous sample populations, participant selection bias, inconsistent assessment methods,

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[1] See *Appendix A* for a glossary of key terms related to psychedelic research and substances used throughout this paper.

[2] MDMA it is frequently differentiated from classic serotonergic psychedelics (e.g. psilocybin, LSD, mescaline and DMT) based on divergences that include mechanisms of action, primary effects, tolerance, risk and dependence. For ease of discussion however, this paper will use the term ‘psychedelic’ as a collective noun that encapsulates MDMA and ‘classical psychedelic’ when MDMA is not included.

and a lack of extended follow-up periods (Aday et al., 2022; Andersen et al., 2021; Breeksema et al., 2022; Haijen et al., 2018; Holze et al., 2024; McNamee et al., 2023; Ona et al., 2022; Schlag et al., 2022; van Elk & Fried, 2023; Watford & Masood, 2024). Compounding these concerns is a critical demographic oversight: the underrepresentation of women[3] in psychedelic research (Shadani et al., 2024). This gender imbalance not only limits the generalizability of current findings but also potentially obscures sex-specific acute adverse effects (AAE), further complicating our current understanding of psychedelic safety profiles. This study seeks to address these identified gaps. As psychedelic therapies stand on the brink of widespread clinical application, it is vital to understand the full spectrum of their effects across diverse populations, particularly if there is a risk that potential sex-specific adverse experience may have been overlooked through under-sampling.

The Neglect of Sex in Psychedelic Research

The neglect of sex as a biological variable (SABV) in biomedical research is not uncommon (Barlek et al., 2022). The National Institutes of Health (NIH) and the US Congress attempted to address this disparity in 2015 and 2016 with the introduction of policies that promote female inclusion in both animal and human studies. The impact has been relatively limited, however; a mere 20% of neuropsychopharmacology studies published before March 2020 adequately evaluated sex differences, despite 72% of the papers reviewed identifying significant sex-related variance (Duffy & Epperson, 2022). This sex disparity extends to psychedelic research, where studies often exclude female participants or fail to report cohort sex composition (Shadani et al., 2024). Women typically constitute only one-third of participants in brain imaging studies examining psychedelic effects (Shadani et al., 2024) and female participation in classical psychedelic research that focus specifically on adverse effects are especially low, ranging from only 17% to 26% (Carbonaro et al., 2016; Johnstad, 2021). The need to examine adverse psychedelic experience in women is underscored by

[3] 'Women' and 'female' in the context of this research refers to the biological sex of an individual (i.e. assigned female at birth (AFAB); chromosomal sex determination, XX), not gender identity (self-representation, social and cultural paradigms).

the complex interplay of physiology, psychological dysfunction, and potentially unique pharmacodynamic responses to these substances.

Female Hormones, Psychological Dysfunction and Psychedelic Pharmacology

Sex-specific hormonal transitions throughout a woman's lifespan significantly affect both the pharmacokinetics and pharmacodynamics of substances. Ovarian steroids have been shown to modulate neurotransmitter systems, neural plasticity, cognition, mood and behaviour (Barth et al., 2015; Birzniece et al., 2002; McEwan & Milner, 2017). These hormonal factors also contribute to sex-based disparities in psychological dysfunction; paradoxically, many of which are being positioned for psychedelic interventions (Kundakovic & Rocks, 2022; Shadani et al., 2024). Women face higher lifetime risks of depression, anxiety disorders, PTSD, and eating disorders (WHO, 2022; American Psychiatric Association, 2017), and increasingly higher rates of substance use disorders (SUD), with evidence to suggest more severe adverse outcomes than men (McHugh et al., 2018). Notably, rapid changes in estradiol levels are linked to perimenopausal depression (Schmidt & Rubinow, 2009), while menopause-related estrogen disruption can dysregulate 5-HT_{2A} receptors and brain-derived neurotrophic factor (BDNF) signalling, predisposing the brain to depression (Chhibber et al., 2017).

The pharmacodynamics of psilocybin and MDMA underscore the relevance of considering sex-specific adverse effects. Psilocybin (4-phosphoryloxy-*N,N*-dimethyltryptamine) is a psychoactive tryptamine found in over 200 species of basidiomycetes (fungi), mostly from the genus *Psilocybe*, and is considered a classical psychedelic (Nichols, 2020). It exerts effects predominantly through activation of serotonin 5-HT_{2A} receptors, inducing profound dose-dependent alterations in emotion, perception, and cognition (Passie et al., 2002; Holze et al., 2020). In contrast, MDMA (3,4-Methylenedioxymethamphetamine), colloquially known as 'ecstasy', 'molly' or ‘E’, is a phenethylamine compound that acts as an indirect sympathomimetic, primarily affecting human monoamine transporters (Hysek et al., 2011; Verrico et al., 2006). It is frequently referred to as either an entactogen or empathogen, based on the profound sense of connection it induces, encompassing both intrapersonal (self-awareness and introspection) and interpersonal (empathy, openness, and social bonding) dimensions (Stocker & Liechti, 2024). While the mechanism of MDMA differs from psilocybin, its subjective effects also largely stem from its influence on the serotonergic system (Sarparast et al., 2022). This serotonergic influence is particularly relevant given that hormonal changes have been shown to affect selective serotonin reuptake inhibitors (SSRI) responses, as evidenced by improved efficacy following hormone replacement therapy in postmenopausal women (Shadani et al., 2024). Consequently, hormonal fluctuations associated with menstruation and developmental life stages may differentially modulate women's experiences with these substances.

Preliminary evidence from studies incorporating SABV appears to support this hypothesis. In a study of healthy, psychedelic-naive volunteers, women reported more intense psychoactive and adverse effects in response to MDMA, including heightened perceptual changes, thought disturbances, and fear of losing body control, compared to men (Liechti et al., 2001). Notably, equivalent doses per kilogram of body weight produced stronger responses in women. Additionally, an animal study using rats has demonstrated that lower doses of psilocybin produce sex-specific effects on fear extinction: accelerating extinction and promoting long-term fear reduction in males, while slowing extinction and increasing long-term fear in females (Miller et al., 2024). These findings suggest that psychedelic-induced physiological changes may not be uniform across sexes and underscores the need for further research with a sex-specific focus.

The Current Study

This study is primarily exploratory. It seeks to addresses the knowledge gap regarding sex-specific effects in psychedelic studies by examining AAE that women report during negative experiences with MDMA or psilocybin. The complex interplay between sex hormones, psychological vulnerability, and psychedelic pharmacodynamics, coupled with the underrepresentation of women in psychedelic research, underscores the importance of this investigation.

Methodological Approach

A qualitative approach was deemed the most epistemologically appropriate. Recent concerns have been raised about the limitations of standardized questionnaires and quantitative measures in assessing psychedelic interventions, citing issues of heterogeneity, inadequate construct validity, and demand bias (Bzdok et al., 2024; Hovmand et al., 2024). Critically, they frequently neglect the negative aspects of psychedelic experience and may be failing to fully capture the complex spectrum of psychedelically induced cognitive and sensory changes (Bzdok et al., 2024; see Hovmand et al., 2024 for a review). Consequently, researchers are increasingly advocating for more qualitative and mixed method approaches in psychedelic research (Haijen et al., 2018; McCartney et al., 2023). Notably, subjective psychedelic experiences have been shown to encompass a broader thematic range than clinical trial data typically capture (Johnstad, 2021) and women who experienced harm during clinical trials have actively called for more phenomenological-based research to identify overlooked risks and adverse events (McNamee et al., 2023; Duhaime-Ross, 2024).

Rather than seeking a singular, objective truth that can be universally extracted and applied, or testing a specific hypothesis or theory, this study employs reflexive thematic analysis of self-reported ‘bad trips’ to provide a more expansive understanding of the negative effects that mediate psychedelic experience for women. Concerns about recall bias are valid - given the potential for memory retrieval errors, cognitive biases, personality traits, external influences, and medical comorbidities to affect recall abilities (Bzdok et al., 2024), however this research recognizes participant recollection and personal interpretation as an inherently valid representation of reality, providing a rich source of insight into an underexamined sphere of psychedelic phenomenology.

The use of large language models (LLMs) was incorporated alongside traditional qualitative methods to enhance analytical insight and rigor. Recent studies have demonstrated promising applications of LLMs in qualitative analysis, including deductive coding of large-scale datasets (Xiao et al., 2023), as a support tool, integrated into collaborative analysis workflows (Gao et al., 2023), and exploration of automated analysis (Byun et al., 2023). LLMs have also shown success in the structured content analysis of free-form psychedelic experience reports (Bzdok et al., 2024). In alignment with Xiao et al. (2023) this study will employ LLMs as an assistive tool for systematic coding. The aim is to not only enhance the coding process, but to also critically evaluate LLMs as qualitative research tools, contributing to the ongoing discourse surrounding the viable integration of AI into qualitative methodology.

The employment of existing self-reports allowed circumvention of the legal and ethical violations inherent with the administration of controlled substances and inducing negative psychological states in human participants. Preclinical research has shown that the selective effects of MDMA on 5-HT neurons are species dependent (O'Callaghan and Miller 1994; Xie et al. 2004), with primates being more sensitive to the neurotoxic effects of MDMA than rodents (Ricaurte et al., 2000), complicating the application of translational studies to human populations. It also addresses another key limitation of animal models, which are unable to capture the nuanced, subjective experience central to psychedelic phenomenology.

Key Definitions and Focus

AAE within the context of this study is defined as any transient effect or event that was perceived as negative, harmful or unwanted, and occurring during the active psychedelic session following substance consumption. These are distinct from negative outcomes, which are delineated as any negative, harmful or unwanted event or effect emergent after the session had terminated (as defined by the participant).

A focus on psilocybin and MDMA experience was determined based on the imminent therapeutic application of these substances for psychological dysfunctions, such as depression and PTSD, which disproportionately affect women (WHO, 2022). Exploring both substances under a single research question enables valuable cross-comparison of their unique effects, particularly given the limited qualitative research comparing subjective psychedelic accounts within a single study (McCartney et al., 2023).

Psilocybin is generally considered safe, with low chronic toxicity and abuse potential (Tylš et al., 2014), however it can cause transient adverse effects. Physical effects may include headaches, fatigue, nausea, vomiting and hypertension (Breeksema et al., 2022; Hinkle et al., 2024). Psycho-emotional effects can include intensified emotions such as anxiety, fear, confusion, mood depression and paranoia (Breeksema et al., 2022; Hinkle et al., 2024). Individuals may also experience distressing cognitive changes, including increased introspection, concentration or memory difficulties, hallucinations and altered perceptions of time or self (Gill et al., 2020; Hinkle et al., 202). In contrast, MDMA poses higher risks of acute toxicity, abuse, and fatality (Gill et al., 2020; Rigg & Sharp, 2018). Common acute adverse effects of MDMA include fatigue, insomnia, headaches, sweating, temperature dysregulation, cardiovascular changes, dehydration, nausea, vomiting, bruxism, and decreased appetite (Breeksema et al., 2022; Gill et al., 2020; Sarparast et al., 2022). Additionally, MDMA can also induce negative psycho-emotional states, including anxiety, fear, confusion, and difficulty concentrating (Breeksema et al., 2022; Sarparast et al., 2022).

Research Questions

In a field driven by pioneering enthusiasm, the imperative for researchers and clinicians to ensure full transparency of adverse psychedelic experiences is paramount (Breeksema et al., 2022). Mental health professionals currently view psychedelic-assisted therapy (PAT) cautiously, hampered by limited knowledge of the substances and processes involved, with education frequently proposed or requested to reduce stigma and improve understanding (Berger & Fitzgerald, 2023; Davis et al., 2021). While acknowledging that AAE cannot be eliminated entirely from pharmacological interventions, this study aims to deepen understanding of psilocybin and MDMA experiences by elucidating AAE reported specifically by women during negative encounters with these substances. By centring analysis around female narratives only, it eliminates the potential confounding influence of male data, enhancing our understanding of sex-specific adverse effects. This research seeks to provide insights for psychoeducation and inform approaches to treatment, risk mitigation, and integration strategies, ultimately enhancing the safety and efficacy of both psilocybin- and MDMA-assisted therapies for women. The analysis was guided by two research questions:

1. What AAE do women report to have experienced during a self-identified 'bad trip' on psilocybin or MDMA?
2. How do LLMs compare to a human coder in assigning predetermined AAE codes to psilocybin or MDMA experience reports?

Method

Participants

First-person, self-reported written accounts were obtained from “experience vaults” on Erowid.org (2024), a publicly accessible online platform. Erowid is a non-profit educational organisation that seeks to provide information regarding psychoactive chemicals and plants, including experimental accounts of psychoactive substance use. The Erowid database is the largest resource of its kind (Bzdok et al., 2024) and has emerged as an invaluable source of experiential information enabling the qualitative examination of well-known and novel drug effects in illicit compounds (Newman et al., 2016; Swogger et al., 2015). Given this established reputation, Erowid was deemed both an appropriate and reliable resource for the current study. Participant sex was the only consistent and reliably reported demographic information in all reports. Ethical approval to use the datasets was granted by the Victoria University School of Psychology Human Ethics Committee (0000031472).

Procedure

Reports were sourced between March and June 2024 using convenience sampling. See *Table 1* for report inclusion and exclusion criteria. To best capture the potential range and diversity of

Table 1

Applied Selection Criteria for Reports Included in Analysis

Inclusion criteria	Exclusion criteria
Reviewed and edited by Erowid.[a]	Unreviewed by Erowid or a 'cellar' report.[a]
Written in English.	Written in a language other than English.
Author self-identified as a woman.	Self-identified as male, intersex or sex not disclosed.
Experience self-reported as negative.	Reports that were not self-identified as negative.
Reports from 2000 – present.[b]	Reports before 2000.[b]
	Polysubstance use e.g. cannabis, alcohol, amphetamine, LSD, cocaine.
	Indication of medically prescribed psychotropics e.g. anti-anxiety agents, antipsychotics, antidepressants (including SSRI, SNRI or MAOI), mood stabilizers and stimulants.
	Indication of a concurrent substance dependence.
	Detailing another individual's adverse experience.

Note. SSRI = Selective serotonin reuptake inhibitor; SNRI = Serotonin-norepinephrine reuptake inhibitor; MAOI = Monoamine oxidase inhibitor.

[a] Reviewed and edited Erowid reports have been approved by at least two trained reviewers and may have undergone minor edits. Unreviewed reports have not been through this process. 'Cellar' reports contain potentially useful data but are deemed unfit for public display due to issues with quality or credibility.

[b] Date range was determined by the earliest report that met all other inclusion criteria.

adverse experience in women, criteria prioritized inclusivity while eliminating potential confounding variables such as the effects of polysubstance use (both recreational or medically prescribed) or concurrent chronic substance abuse. These reports were excluded to avoid conflating effects that may be attributable to other substances or atypical drug interactions. Cannabis is often used in conjunction with MDMA and psilocybin to intensify experiences or mitigate negative effects (Hunt et al., 2009; Kuc et al., 2022; Licht et al., 2012) but this combination can also cause adverse reactions (Carbonaro et al., 2016; Palamar et al., 2016) and was implicated in 37% of psilocybin-related cases seeking emergency medical treatments in 2017 (Kopra et al., 2022). Given the reliance on self-reported experiences, this approach provided the best possible means of isolating and examining adverse effects specifically attributable to MDMA or psilocybin. It also aligns with therapeutic applications which typically involve single-substance administration.

A separate search for each substance was conducted on Erowid in the following categories: "Difficult experiences", "Bad Trips" and "Train Wrecks & Trip Disasters". Once selected for analysis, the anonymized reports were assigned ID numbers to allow discernment of drug type. Final analysis included fifty-six reports (27 MDMA and 29 psilocybin). Psilocybin reports averaged 1,598 words, while MDMA reports averaged 690 words. Overall, average report length was 1,161 words. To provide contextual information about the dataset, participant, environmental and experience characteristics were extracted from the final report collection where possible, including but not limited to age, weight, dose, relationship and employment status, pre-existing mental health conditions, experience preparation and intent, social context, location where most AAE were experienced, and if medical assistance was required.

Materials

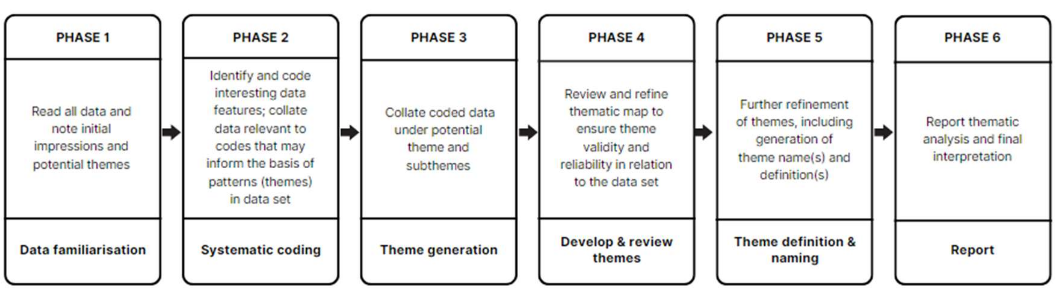
Reports were copied into individual word documents and uploaded to NVivo 14 (Lumivero, 2023), a software package designed for qualitative and mixed-methods research. NVivo was used to highlight relevant data and conceptualise themes. Two LLMs - Claude 3.5 Sonnet (Anthropic, 2024) and ChatGPT-4 (OpenAI, 2024) were utilized to enhance the analysis process through periodic reviews of code application and assessment of intercoder reliability. Substance quality and composition were unverifiable as participants independently obtained and determined their own doses.

Design

Experience reports were thematically analysed in six phases in accordance with an adaptation of Braun and Clarke's (2006, 2020) framework (see *Figure 1*), chosen for the flexibility it allows when a research question encompasses both exploratory and confirmatory approaches (Braun & Clarke, 2006). To facilitate as comprehensive identification of AAE as possible, an exploratory approach was principally employed, however analysis was also partially confirmatory, in that it was undertaken with some anticipation of "set" and "setting" factors (Hartogsohn, 2017) as well as reported AAE from previous studies (Breeksema et al., 2022; Carbonaro et al., 2016; Hinkle et al., 2024; Liechti et al., 2001). A realist paradigm was adopted, whereby it was assumed language directly reflects participant experience and meaning.

Figure 1

An Illustration of the Thematic Analysis Undertaken for Theme Development



Note. Adapted from Braun and Clarke (2006, 2020).

Phase 1 involved data and narrative familiarisation through multiple readings. Preliminary code generation in Phase 2 was undertaken using a dual deductive-inductive process (Braun and Clarke, 2006). Semantic and latent coding was employed with no prioritisation given to either i.e. codes were produced when meaningful information of either kind was interpreted. Any item could therefore be double coded in alignment with meaning supplied by the participant and researcher interpretation (Braun & Clarke, 2020). This aligns with the theoretical assumptions of the analysis and the approach argued for by Braun and Clarke (2019, 2020, 2022), which positions the researcher as central, actively engaging in the interpretation of codes and themes and determining their relevance to the research question. As previously indicated, some codes were based on the data while taking into consideration set and setting, as well as previously identified AAE. This enabled identification of specific AAE during Phase 2 as part of the analysis to facilitate a comparative discussion of AAE reported in clinical trial data. Codes were grouped according to thematic similarity in Phase 3. A review of all themes and sub-themes was conducted in Phase 4, consolidating or eliminating redundancies allowing identification of clear and comparable concepts. In Phase 5, definitions for each theme and subtheme were finalised and named before reporting the final interpretations in Phase 6.

To provide a supplementary perspective on code application, Claude 3.5 Sonnet and GPT-4 were independently employed at the end of Phase 2. Using a framework for LLM-suggested modelling under human review (Eschrich & Sterman, 2024) and following a deductive coding process, both models were tasked with proposing qualitative codes for a randomly sampled subset of reports (ten per substance) based on the researcher-developed codebook. Researcher oversight was maintained through critical evaluation and sense-checking of code suggestions, comparing them with the report and researcher coding. Where appropriate, codes may have been transposed, additional codes applied to a report or rejected. The common scenarios in which LLM generated codes were disregarded, included:

a) Irrelevance, the code did not accurately reflect any content in the report;

b) Misinterpretation, details or nuance in the text was misunderstood;

c) Over-interpretation, whereby codes of inference were not applicable to the text or

d) Redundancy, the suggested code was better captured by another code that was already assigned and more appropriate.

This approach balanced LLM-assisted analysis with human interpretation and reflexivity, aiming to enhance coding robustness while providing a comparative test case for LLM application as an assistive tool in qualitative analysis. Peer debriefing sessions with a supervisor and a peer research group were also conducted weekly throughout the analysis.

Results

LLM-Assisted Code Review

The average pairwise agreement of the Phase 2 LLM-assisted codebook review was 36%. Claude demonstrated 46% agreement with researcher coding across both substances, compared to 28% for ChatGPT. Complete agreement across all three coders occurred in 16% of reports. Overall coder agreement was the same for both MDMA and psilocybin reports (33% average pairwise, 13% average agreement). For MDMA reports researcher-Claude achieved highest agreement (52%), followed by Claude-ChatGPT (37%) and researcher-ChatGPT (33%). For psilocybin, agreement was highest between researcher-Claude (39%), followed by Claude-ChatGPT (30%) and researcher-ChatGPT (23%). Claude consistently assigned more codes than ChatGPT but on average less than the researcher. For both substances, ChatGPT assigned the least number of codes (M = 8). The lowest overall agreement (5%) was observed in a 1,302-word psilocybin report, while the highest (70%) was in a 160-word MDMA report. For more detailed results of this process, see *Appendix B*.

Participant Characteristics

Details pertaining to participant characteristics can be found in *Appendix C*. Most participants did not disclose their age (75%, n = 42), relationship status (60.7%, n = 34), employment status (69.6%, n = 39), or a mental health condition (91.1%, n = 51). Mean age was 19 years (SD =

7.3, range 16–45). Participant body weight averaged 56kg (*SD* = 11.8, range 40-113). The most frequently disclosed relationship status was partnership with a male (33.9%, *n* = 19) employment status as "student" (17.9%, *n* = 10), and 8.9% (*n* = 5) reported having pre-existing depression. No participant reported a racial or ethnic identity.

The Adverse Psychedelic Experience

Information regarding participant dose, set and setting can be found in *Appendix* D. For MDMA, the most common participant dose was 1 tablet (44.4%, *n* = 12), while psilocybin users typically consumed a strong dose of 2.5g – 5g (48.3%, *n* = 14). Recreational use was the primary intent, accounting for 64.3% (*n* = 36) overall and 92.6% (*n* = 25) of MDMA users specifically. For 26.8% (*n* = 15) of participants this was their first experience with the substance, while 57.1% (*n* = 32) had used the substance before. Consumption typically occurred in social settings, with 50% (*n* = 28) using the substance among friends and 26.8% (*n* = 15) with both partner and friends. Adverse effects were predominantly experienced in familiar environments such as at home or that of a friend (57.1%, *n* = 32), especially for psilocybin participants (75.9%, *n* = 22). In contrast, MDMA users more often reported AAE in public locations such as musical events or parties (44.4%, *n* = 12). One psilocybin participant (1.8%) indicated they were menstruating. A small portion of participants sought medical assistance for AAE (8.9%, *n* = 5), with one MDMA participant (1.8%) requiring prolonged hospitalization.

Themes

Analysis generated four themes of female adverse MDMA and psilocybin experience: *Overwhelming Affect (OA), Amplified Physiological Awareness (APA), Dissonant Altered States (DAS)*, and *Dynamic Influences (DI)*, as well as thirteen associated subthemes (see *Table 2*). It is important to acknowledge that while discrete themes have been purposefully defined to describe the distinctive facets of AAE, this is primarily a conceptualisation employed to facilitate discussion of findings. Themes were frequently intrinsically interconnected components of a holistic experience, that may converge and reinforce one another.

Table 2

Qualitative Responses Describing the Acute Adverse Effects of MDMA and Psilocybin Experiences Reported by Participants Organised by Theme and Subtheme

Theme and subthemes	Description	Exemplars
Overwhelming Affect		
Emotional distress	Emotional responses characterised by negative valence. Primarily manifestations of anxiety (from nervous worry to panic), and fear (from unease to terror). Feelings of profound confusion, sadness and despair were also reported. These emotional responses contributed to an overriding sense of discomfort and distress.	*"I sobbed for hours, talking a lot of gibberish, trying to explain the overwhelming emotions. Never have I been so completely sad, or felt so unintelligent, or been so frightened of what was happening."* ("*Mouse*" *Report ID: 72,630*) *"This may not sound that bad, but it was, I can't even express it. Fear locked onto my brain, my body couldn't stop shaking (even now it's still shaking). I thought I was going to lose my mind, suffocate, and freeze to death simultaneously."* ("*PhetamineD*" *Report ID: 39,888*) *"I had that panicky, ominous feeling when you know you're about to go into a bad trip and hell is about to break loose."* ("*dashboardm3lted*" *Report ID: 85,098*)
Existential distress	Profound and frightening thoughts of an evil or malevolent presence, irreversible personal harm, or mortality. Includes experiences of death certainty, medical	*"I come to the conclusion that I have taken it too far and am forever imprisoned in the terrible crevices of insanity that have always existed in my mind." ("Julinfinity" Report ID: 68,977)* *"That eminem song popped into my mind, and all I could think about was that the girl ended up dying. *I* was dying. I cried, and screamed, and became*

	intervention or suicidal ideation.	*engulfed with fear. I didn't want to die."* (*"PinkFlower" Report ID: 50,799*) *"The room around me became dark and overwhelmingly evil. Everywhere I looked there was nothing to turn to for comfort and it seemed as if all of my possessions were turning on me and reminding me of how bad of a person I am." ("Porridge" Report ID: 95,189)*
Isolation and vulnerability	Sense of disconnection to self and others. Includes, vulnerability, helplessness and cosmic alienation. Can involve perceived hostile environments and a lack of help seeking or obtaining abilities. Also includes feelings of intense self-doubt and despair.	*"A feeling of hopelessness surrounded and crashed on me like a wave. I couldn't escape my faults coming down on me, one after another after another. After another. I sometimes wimpered 'Please stop, this has got to stop.' This room had become my own awful prison cell. I was scared and alone." ("MissEcho" Report ID: 44,519)* *"No one would stay on the phone with me for longer than 5 minutes, everyone was hanging up, I felt as if the world hated me, and that the house was capturing me and telling me to stay inside and not socialize." ("Jessi" Report ID: 56,307)* *"I felt like I might just be God, trying to distract myself from my loneliness by creating this pattern of life and death on Earth…" ("KanonBosatsu" Report ID: 90,298)*
Loss of control and disorientation	Primarily feelings of powerlessness, confusion, and AAE inescapability. Often associated with fight or flight responses. Can also involve involuntary behaviours, cognitive struggles, and rapid shifts in perception.	*"I couldn't stop arching my spine and had to have every muscle in my body flex---my hands, back, feet, neck, spine, anything. I was so scared in my head, but I could not control anything that was happening to my body. I couldn't do anything, and it terrified me." ("Anna" Report ID: 48,974)* *"I kept kicking and screaming and banging my body anywhere. Hitting whoever came near me and screaming at them for talking to me."* ("*Melisssssa*" *Report ID: 23,884*)
Negative self-perception	Negative affect linked to critical self-evaluation,	*"I was alone. My grades were bad. I was being so weird. I could never be a good person. Squeezing*

	including sadness, guilt, shame, and worthlessness. Often triggered by distorted self-image, past trauma, or unresolved issues. Contemplating pharmacodynamic responses could lead to social withdrawal and rejection of support, anticipating criticism or feeling burdensome.	*my eyes closed I sobbed without crying. Desperation, I had to stop feeling this way. I felt emotionally pained and feeling like I had screwed myself out of ever being a good person. I had disappointed myself and countless others." ("MissEcho" Report ID: 44,519)* *"…my eyes were so swollen I looked like I had an allergic reaction and my body was so bony and gross and hunched over, I looked like a monster." ("Garden Witch" Report ID: 116,399)* *"I felt extremely guilty towards my friends because I felt like I was preventing them from enjoying themselves so I told them just to leave me alone and to forget about me…" ("Sab" Report ID: 27,911)*
Amplified Physiological Awareness		
Hypersensitivity to normal physiology	Heightened awareness transformed normal bodily functions into intense, alarming experiences. This hypersensitivity to physiological change could cause significant experiential discomfort.	"*I started to sweat, then feel cold, then sweat, then feel cold."* ("*Psilocybin*" *Report ID: 35,133*) *"…then suddenly it hit me really hard, and really fast. I could feel my heart absolutely racing, which I knew was normal, but I felt funny. It felt somehow different to me…"* ("*Carley" Report ID: 35,133*)
Distressing physiological responses	Physical discomfort beyond heightened awareness, constituting more severe bodily distress that would be disconcerting or upsetting irrespective of intoxication status. These were often reported in conjunction with and therefore exacerbated psychological distress	*"After about half an hour I started feeling sick to my stomach and had a bad headache."* ("*Bionaire" Report ID: 933)* "*The throwing up was painful, nothing like what I've felt before and eventually it was all just dry painful heaves. It was uncontrollable and just extremly painful and lasted a long time. It took about 4 hours of sitting in the bathroom on and off puking for the high to eventually come down*." ("*Sheena" Report ID: 28,506*)

	developing a reinforcing adverse mind-body experience.	
Ineffable physiological experience	Unusual, novel or indescribable bodily sensations that defied participant description, understanding and often challenged experiential control. Could led to confusion, anxiety and feelings of helplessness or panic.	*"I saw the look on his face and tried to explain, but neither he, nor I could understand what I was saying. The words were perfect in my head, but when they came out, they were backwards, wrong, or not even real words."* (*"Quack" Report ID: 56,521*) *"My lower jaw was flapping very quickly, independently of my best efforts, and I kept trying to grab it and keep it still."* (*"swingsetdreams" Report ID: 106,197*) *"I couldnt begin to describe what this felt like other than it was like things inside me were being ripped and torn, almost like something wanting to leave the inside of me and that was the only way to do it"* (*"Sheena" Report ID: 28,506*)
Dissonant altered states		
Cognitive fragmentation	Unsettling disruption to normal cognitive function and agency. Often experienced as racing thoughts, incoherence and memory issues. Associated with perceptual changes which led to cyclical thinking and fears of insanity.	"*My thoughts began moving so fast through my head that I couldn't understand them anymore"* (*"dashboardm3lted" Report ID: 85,098*) "*I couldn't think, I couldn't reason, I couldn't remember from one minute to the next. I doubt I could have counted for ten paces before forgetting what number I was up to. I am not exaggerating, it was that bad*" (*"Inanna" Report ID: 62,728)* "*I was trying to pull myself together to no avail. I wanted to do something, anything to make it stop--break a window or the bathroom mirror, slit my wrists-anything to make this madness stop and get away from the horrible, incapacitating thoughts that were in my head"* (*"Anonymous09" Report ID: 14,308*)
Perceptual fragmentation	Visual, auditory, or tactile distortions that created a	"*The hallucinations were fucked up by now. Scary almost. Everything ran together, objects rushed at*

	sense the world was changing in insensible and threatening ways, leading to disorientation and distress. This could include altered feelings of embodiment, disembodiment, depersonalisation and derealisation.	*me. Sounds overwhelmed my thoughts and I couldnt hear things. It sounded like I was in the middle of a million people all talking at the top of their lungs not being able to understand any one of them*" (*"Sheena" Report ID: 28,506*) "*Mushrooms seemed to have fried my ability to perceive reality from fantasy. I was unable to recognize the fine line between the real world and the fantasy world that only existed in a fragment of my imagination. What wasn't real, what was real, I could not tell*" (*"Jenny" Report ID: 26,710*) "*My reflection in the mirror seemed strange and foreign. I didn't understand why my reflection was doing everything I was doing and I didn't recognize myself visually*" (*"Fox and Rabbit*" *Report ID: 19,907*)
Dynamic Influences		
Environmental	Ambient conditions, events and location could provoke AAE, especially when associated with change. Loud, bright, or crowded settings caused feelings of overwhelm. Temperature extremes triggered discomfort. Familiar spaces could become unsettling or hostile, transforming ordinary environments into sources of distress.	"*Finally we got there but it was hot, there was little shade, we were all feeling uncomfortable from our stomachs"* ("*Mouse*" *Report ID: 72,630*) *"I was sitting on my bed, looking at the drips and spoon and I thought to myself 'Damn, this room is so messy', but I guess I said it out loud because my friend said 'Yeah, it's like a prison cell, haha.' Just my thoughts exactly. I needed to be out of the dark, messy, prison cell room because it was evil." ("MissEcho" Report ID: 44,519)* "*It was in a enclosed building and I very soon started to panic and feel closed in. Once the opening band came on I felt like I couldnt breath, I was gettin horrible chest pains and could not relax."* ("*eyesoftheworld" Report ID: 75,292*)
Interpersonal	A heightened social awareness during interactions that frequently	*"I felt as if everyone at the party was giving me dirty looks, and I thought everyone was whispering about me. Also, although I knew every single*

	compounded AAE, prompting self-consciousness, anxiety, and paranoia. Also includes concerns of external judgement; fear of discovery by sober individuals and distorted perceptions of others intensified discomfort. Concerns about loved ones' reactions prompted guilt and shame.	*person at the party, I felt like they were strangers, and I began to get extremely paranoid" ("Rollerr" Report ID: 78,637)* *"…everyone came in the room to see what was wrong with me. I freaked out, I was so scared because I didn't know who anyone was" ("Anna" Report ID: 48,974)* *"I could hear my husband talking to his online friends in the next room, I was really worried he would get up and see me in that state, eyes and face swollen, lying naked on the floor, I honestly felt so ashamed and how could I let myself get to that point?" ("Garden Witch" Report ID: 116,399)*
Intrapersonal	Destabilizing effects of unpredictable fluctuations in internal state that contributed to experience adversity. This includes rapid onset, emotional instability, and expectation-reality mismatch which could also sometimes lead to impulsive redosing.	*"It seemed that my emotions would change every few minutes from sad to frustrated to scared."* (*"sara" Report ID: 20,417*) *"i didnt feel anything at all for like an hour and i was like this is stupid. So i was watching tv and it just hit me... really hard!"* (*"Trina" Report ID: 5,954*) "*I began to slide on his bathmat, and soon was on my back. What the hell was going on? Why was I on my back? I soon realised what had happened, and burst into hysterical laughter, and sat back up. The laughter snapped back into hysterical crying. What was happenening? Was I dying? And then it happened once more, I was on my back again. I burst back into hysterical laughter at how rediculous it all was. I propped myself back up, and broke into tears once more"* (*"PinkFlower" Report ID: 50,799*)

Overwhelming Affect

OA describes the adverse psycho-emotional effects characterized by intense, unexpected or reactive feelings of confusion, anxiety, fear and sadness. The dominant states of reported experience can be understood as intense, frequently amplified by a perceived inability to self-regulate. Five subthemes elucidate the nature of these psycho-emotional effects.

Emotional Distress. Describes the spectrum of negatively valanced emotions that caused psychological distress. These were predominantly characterised by forms of anxiety and fear that spanned from mild unease to abject terror. Rapidly shifting emotions were common, often in response to effects or events described in the other themes which contributed to a sense of instability or physiological panic. Self-reflection prompted sudden and intense sadness or depressive rumination in some participants. Many individuals grappled with both the nature and inability to comprehend the drug influences on their body or altered state. The distressing nature of phenomenology frequently led to a strong conviction of dying or impending death, central to the subtheme *Existential distress*.

Existential Distress. Encapsulates the disturbing phenomenology grounded in the fear of irreversible damage to self, both physically and mentally. Several participants were afraid to fall asleep believing they would not wake up. This distress encouraged thoughts of needing medical assistance or a desperate desire to return to normalcy. Many expressed a regret for taking the substance in the first place and were desperate for the experience to end, one even reporting suicidal ideation to escape the perceived turmoil of their experience. Some participants experienced a sense of impending doom, or preoccupation with the impression of a malevolent presence or entities.

Isolation and Vulnerability. Some participants described a profound sense of isolation, even when physically surrounded by others. This disconnection could extend to a cosmic scale in a few instances, particularly when combined with altered perceptions of embodiment, derealisation and ego dissolution. Many experienced this state as a pervasive sense of exposure, leading to feelings of defencelessness exacerbated by a perceived inability to control their experience, themselves or seek help effectively. These states appeared to trigger a cascade of negative psychological effects, including helplessness, loss of meaning, and nihilistic thinking. The environment could be perceived as hostile, intensifying feelings of vulnerability. Furthermore, this condition often prompted severe self-doubt and self-criticism.

Negative Self-Perception. Experience within this subtheme was associated with states of anxiety, sadness, and self-conscious emotions such as guilt, shame, and embarrassment. Some participants were confronted both intentionally and unintentionally, by deeply personal concerns, fears or insecurities. Often this involved revisiting past behaviours, trauma, unresolved issues or existential questions which provoked self-critical internal dialogue and feelings of worthlessness, insignificance or despair. These feelings were intensified by negative self-appraisal or perceptual distortions related to personal physical appearance. Shame and embarrassment were also frequently expressed in relation to their negative pharmacodynamic responses, especially in social contexts. They worried how others might perceive their drug-induced behaviours and feared becoming a burden to those around them. In some cases, this led to self-punishing behaviours, such as refusing socially offered comfort or support. The perceived disappointment from loved ones was a commonly feared consequence.

Loss of Control and Disorientation. This was a pervasive subtheme across all experiences. Participants frequently reported a profound sense of powerlessness to alter or control their state, which intensified the frightening nature of the experience. This feeling of AAE being inescapable was often intertwined with confusion, panic, and fear, encouraging flight or fight responses. Sometimes this manifested as feeling trapped or a desperate desire to escape the experience. Loss of control manifested in three ways: physical, cognitive and behavioural. Participants often described uncontrollable body movements, falling, flailing, or an inability to coordinate their actions, as described in the *APA* theme. Many also struggled to control their thoughts or responses to sensory input and information such as hallucinations, as outlined in *DAS*. A small subset of participants reported engagement of uncharacteristically aggressive and volatile behaviour as outward expressions of their internal distress, which involved actions that were harmful or destructive to self, others or property and required forms of restraint. These behaviours could include clawing at skin, hair pulling, yelling, hitting and kicking others, and throwing items. The *DI* theme highlights how rapid shifts in AAE contributed significantly to overall feelings of disorientation and loss of control.

Amplified Physiological Awareness

This theme encompasses the influential relationship between heightened awareness and bodily sensations that simultaneously induced and exacerbated physiological and psycho-emotional distress. These somatic changes often felt unpredictable and uncontrollable, contributing to the overall intensity and disorientation of the experience. The marked salience of these adverse physiological effects frequently led to a corporeal preoccupation that dominated the narrative. Three distinct yet interrelated subthemes were identified, representing the spectrum of physiological effects that could initiate or contribute to a compounding cycle of distress.

Hypersensitivity to Normal Physiology. Can be understood as the intensification of bodily processes that under normal circumstances might go unnoticed or cause little to no concern but became a source of frustration, alarm or anxiety, even in the absence of concrete harm. Key characteristics included bodily temperature changes (hot, cold and oscillating, including excessive sweating), mydriasis (pupil dilation), hyperactivity and tachycardia (rapid heartbeat).

Distressing Physiological Responses. Describes physiological discomfort or altered functioning that is distressing irrespective of intoxication, but in conjunction with substance amplification, exacerbated a profound sense of overwhelming mind-body distress. The most recurrent AAE in this subtheme were gastrointestinal disturbances including both excessive or prolonged nausea and/or vomiting.

Ineffable Physiological Experience. Atypical bodily sensations characterized by perceived novelty, unusualness, or descriptive difficulty. The inability to adequately comprehend these sensations or reclaim control over them challenged bodily and therefore experiential understanding. Much like the other two subthemes of *APA* these could elicit frustration, confusion, alarm or anxiety and lead to feelings of fatigue, helplessness or panic depending on their nature, intensity and duration. Bruxism, paresthesias and speech impairments were some of the more noted effects from this subtheme.

Dissonance of Altered States

This theme encompasses the perceptual and cognitive alterations that destabilized participants normative conceptualizations of self and reality, and directly contributed to adversity through a sense of uncontrollable and distressing discombobulation. These states were characterised by two subthemes: *perceptual fragmentation* and *cognitive fragmentation*.

Perceptual Fragmentation. Describes the visual, auditory, and tactile distortions that created a sense of the world changing in insensible and threatening ways. Participants grappled with altered feelings of embodiment, disembodiment (disconnection from body), and depersonalization (disconnection from self). Participants also experienced instances of derealization (disconnection from reality), including disconcerting temporal misperceptions where time seemed lost or significantly slowed down. These perceptual alterations and the inability to distinguish them from reality led to increasing disorientation and fear, often provoking existential anxiety as the delineations between self and the world became indistinct or seemingly non-existent.

Cognitive Fragmentation. Encompasses unsettling disruptions to normal thought processes and cognitive function. While participants could recognise the influence of the substance on their state, they felt unable to regain cognitive control. This manifested as racing or uncontrollable thoughts, difficulty maintaining coherent thinking, and memory disruptions. The inability to comprehend the perceptual changes often resulted in cyclical thinking and feelings of insanity. This subtheme captures a general loss of cognitive agency that contributed to an overall sense of helplessness and fear, compounding the challenges presented by the perceptual and existential shifts described in the *perceptual fragmentation* subtheme.

Dynamic Influences

This theme describes the external and internal factors associated with uncontrollable flux which prompted or contributed to the negative valence of the experience. These correlate with the concepts of set and setting, and dosing influences which have been shown to influence and compound the unpredictable and distressing nature of altered states of consciousness.

Environmental Dynamics. Ambient conditions, contextual events, and changes in location not only triggered AAE but also became adverse events themselves. Environments with loud music, bright lights, and crowded spaces often led to a sense of overwhelm and intolerance. Temperature extremes were similarly associated with negative effects. Notably, even ordinary, familiar, or typically comfortable spaces, such as a personal bedroom, could transform into settings perceived as foreign, hostile, or evil.

Interpersonal Dynamics. Describes a heightened social awareness during interactions that adversely affected participants. Increased self-consciousness and social anxiety frequently fed into a sense of paranoia, which could intensify during interactions with sober individuals. Fear of their intoxicated state being discovered or recognised often resulted in panic, especially when combined with perceptual and cognitive effects that made familiar people appear distorted or unrecognizable. Participants also grappled with how their current state might affect or be perceived by loved ones, such as parents, partners or friends, leading to feelings of guilt and shame. Some reported conflict with partners or friends during the experience, exacerbated by communication difficulties that led to relationship strains. These issues were further complicated if the other party had also consumed an intoxicant.

Intrapersonal Dynamics. The destabilizing effects of unpredictable fluctuations in internal state that contributed to adverse experience. These were characterised by a complex interplay of rapid onset, affective lability, and expectation-reality mismatches. In some instances, an expectation-reality mismatch led to impulsive redosing, "*I was becoming frustrated that the effects were taking so long to show, and I had convinced myself that either the mushrooms were bunk, or I was immune to them. My only thought was to take more. Again, I ingested another .5, completely losing track of how much I was taking in on my first trip"* (*"Jisatsu" Report ID: 016PE*).

Thematic Composition of Adverse Experience by Substance

A two-way mixed ANOVA was conducted to examine substance-specific thematic effects in adverse experiences. There was no significant main effect of substance type on adverse experience, $F(1, 54) = 2.63$, $p = .111$, $\eta^2 = .046$, indicating there was a similar overall prevalence of AAE between MDMA and psilocybin. The observed power was low (.36) however, suggesting caution in interpretation due to increased risk of a Type II error.

A significant main effect of thematic prevalence was found across reported adverse experiences, $F(3, 162) = 14.0$, $p < .001$, $\eta^2 = .206$. Post-hoc Bonferroni-corrected pairwise comparisons revealed the difference between *OA* and *APA* (.16 95% CI [.09, .26]), *DAS* (.15, 95% CI [.07, .23]) and *DI* (0.12 95% CI [.06, .19]) were statistically significant (p <.001). This indicates that *OA* was significantly more prevalent than all other themes. Differences between *APA* and *DAS* (-.01, 95% CI [-.08, .08]), *or DI* (-.03, 95% CI [-.12, .05]), and *DAS* and *DI* (-.03, 95% CI [-.11, .05]) were not significant (p = 1.000).

A significant interaction between theme and substance type was observed, $F(3, 162) = 10.88$, $p < .001$, $\eta^2 = .168$, suggesting different thematic compositions of adverse experiences for MDMA and psilocybin. Post-hoc Bonferroni-corrected pairwise comparisons (see *Table 3*) revealed that *OA* was the most prevalent theme and significantly more prevalent than *DI* for both MDMA and psilocybin (see *Figure 2*).

Table 3

Pairwise Comparisons of Theme Prevalence with Bonferroni Correction by Substance

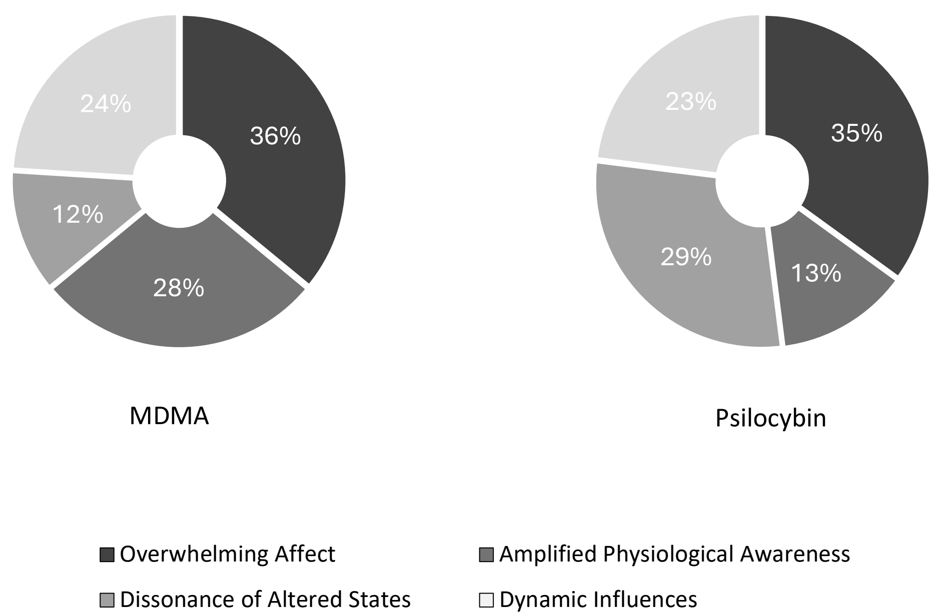
	Theme	Comparative Theme	*Mean diff.*	*SE*	*p*	95% CI	
						LL	*UL*
MDMA	OA	APA	.09	.036	.138	-.01	.19
		DAS	.24***	.040	<.001	.13	.35
		DI	.12**	.035	.007	.02	.22
	APA	DAS	.15**	.042	.003	.04	.27
		DI	.04	.044	1.000	-.09	.16
	DAS	DI	-.12*	.040	.030	-.23	-.01
Psilocybin	OA	APA	.23***	.035	<.001	.13	.32
		DAS	.06	.039	.627	-.04	.17
		DI	.12**	.034	.004	.03	.22
	APA	DAS	-.16**	.040	.001	-.27	-.05
		DI	-.10	.043	.115	-.22	.01
	DAS	DI	.06	.039	.780	-.05	.17

Note. Thematic pairwise comparisons (Bonferroni post-hoc test); OA = Overwhelming Affect, APA = Amplified Physiological Awareness, DAS = Dissonant Altered States, DI = Dynamic Influences; *SE* = Standard Error; CI = confidence interval; *LL* = lower limit; *UL* = upper limit.

*p < .05. ** p < .01. *** p < .001

Figure 2

Average Thematic Composition of Adverse Experience for MDMA and Psilocybin



For MDMA experiences, *DAS* was significantly less prevalent than all other themes. For psilocybin experiences, *APA* was significantly less prevalent than *DAS* and *DI*. Independent t-tests were conducted to compare how the prevalence of each theme differed between MDMA and psilocybin experiences. Levene's test indicated homogeneity of variance for all themes ($p < .05$). Shapiro-Wilk tests revealed violations of normality for some themes, necessitating non-parametric (Mann-Whitney U) analyses (see *Table 4*).

APA was significantly more prevalent in MDMA experiences and *DAS* was significantly more prevalent in psilocybin experiences, both with large effect sizes. No significant differences were found for *OA* or *DI*. Non-parametric tests confirmed these results, addressing violations of normality.

Table 4

Thematic Comparison Between MDMA and Psilocybin

	MDMA		Psilocybin					Mann-Whitney U	
Theme	*M*	*SD*	*M*	*SD*	*t*(54)	*p*	*d*	*z*	*p*
OA	.36	.10	.35	.12	.022	.825	0.060	-.62	.533
APA [a]	.28	.17	.13	.09	4.15	< .001	1.111	-3.97	< .001
DAS [a]	.12	.11	.29	.15	-4.71	< .001	-1.258	-4.05	< .001
DI [a]	.24	.13	.23	.13	0.28	.778	0.076	-.57	.566

Note. *N* = 56, MDMA (*n* = 27) and psilocybin (*n* = 29); OA = Overwhelming Affect, APA = Amplified Physiological Awareness, DAS = Dissonant Altered States, DI = Dynamic Influences.

[a] Mann-Whitney U test is reported because Shapiro-Wilk tests indicated normality violation for this theme.

Substance-specific AAE: frequency analysis

Frequency analysis of specific AAE revealed more nuanced patterns, both supporting and enriching the thematic compositions previously described between substances (see *Appendix E*). F*ear, despair, and distress*, was almost ubiquitously reported, being the most frequent single AAE for both substances (MDMA, 96% and psilocybin, 100%). Higher rates of *panic* (79% versus 48%) and *crying and tearfulness* (52% versus 22%) were reported by psilocybin participants.

Psilocybin users reported perceptual alterations more frequently than MDMA users, with higher reports of *hallucinations* (69% versus 30%), *illusions* (69% versus 15%) and *temporal distortions* (55% versus 19%). Similarly, psilocybin participants also experienced more profound alterations to their sense of reality, with higher rates of *depersonalization* (45% versus 7%) and *derealization, dissociation, or sense of surrealness* (76% versus 37%). Cognitive effects were also markedly different. *Changes to thought speed* (34% versus 15%) and *concentration or memory difficulties* (31% versus 7%) were less prominent in MDMA users.

Substance-specific patterns were also observed in physiological responses with MDMA participants reporting more *bruxism* (37% versus 3%), *tremulousness* (37% versus 14%), and body temperature fluctuations (56% vs 24%). Psilocybin participants reported higher rates of *nausea* (45% versus 30%), *incoordination* (31% versus 26%) and *hyperactivity* (28% versus 11%).

Some effects were similarly prevalent, including *abnormal bodily sensations* (74% versus 76%), *vomiting* (33% versus 34%) and *anxiety* (59% for both) for MDMA and psilocybin, respectively. The two instances of suicidal ideation (7%) were reported by psilocybin participants.

Discussion

This study explored female adverse experiences with MDMA and psilocybin, while also investigating the feasibility of LLMs as an assistive tool for qualitative analysis. Four interrelated themes characterized adverse female psychedelic experience: *Overwhelming Affect, Amplified Physiological Awareness, Dissonance of Altered States* and *Dynamic Influences*. While the overall experiential adversity was similar across both substances, distinct patterns emerged in their thematic composition. *OA* was a predominant theme across both substances, and *DI* was comparably prevalent for all participants. Two themes showed notable substance-specific differences: *APA* was reported more frequently by MDMA users but was the least common in psilocybin experience. Conversely, psilocybin users more frequently reported *DAS*, which was the least prevalent in MDMA reports.

It is crucial to first reemphasise the nature of these substances and their intended application as "experiential medicines" (Mitchell & Anderson, 2024, p. 96). While adverse effects are often considered as discrete phenomena for quantitative purposes, one of the key findings of this study is that AAE manifest not as isolated incidents, but dynamic, interconnected responses that cascade through multiple physiological and psychological domains. While outside the scope of the current study, this observation suggests the need for more comprehensive frameworks in this field, capable of capturing the complex psychophysiological responses to these substances. This approach aligns with calls for more integrative methodologies to better understand the mechanisms of psychedelics, both generally and therapeutically (Carhart-Harris & Friston, 2019; Yaden & Griffiths, 2021).

The analysis of individual AAE largely aligned with existing psychedelic research (Breeksema et al., 2022; Carbonaro et al., 2016; Freye, 2009; Hinkle et al., 2024; Johnstad, 2021; Liechti et al., 2001), but several important variances did emerge. Notable was the absence of headaches - reported in 39.9% of participants in mixed-gender studies (Hinkle et al., 2024) - and the emergence of a novel physiological phenomenon characterized as a *feeling of wetness*, observed across both substances. Most significantly however, the prevalence of commonly associated AAE for both substances was notably higher than previously reported in mixed-gender studies (Breeksema et al., 2022; Hinkle et al., 2024). The near-universal experience of *fear, despair,* and *distress* (98%) substantially exceeded previous reports (27%, Andersen et al., 2021; 69%, Johnstad, 2021). Similarly, psilocybin-associated effects such as *anxiety* (59%), *paranoia* (48%), and *visual hallucinations* (69%) were markedly higher than rates in Hinkle et al.'s (2024) meta-analysis (22.3%, 12.2%, and 46.6% respectively).

The heightened prevalence of adverse effects among women may reflect sex-specific differences in the serotonergic system. Serotonin (5-hydroxytryptamine, 5-HT) influences almost every aspect of human behaviour and physiology, regulating mood, fear, anger, stress, memory, attention, and various physiological processes, from bladder control and bowel motility to cardiovascular function; "it is difficult to find a human behaviour that is not regulated by serotonin" (Berger et al., 2009, p. 2). As previously indicated, both psilocybin and MDMA interact with this system, though through distinct mechanisms. Psilocybin selectively targets multiple serotonin receptor subtypes ($5HT_1$, $5HT_4$, $5HT_5$, $5HT_6$, $5HT_7$), showing especial affinity for the $5\text{-}HT_{2A}$ receptor (Nichols, 2004; Tylš et al., 2014), while MDMA acts as a potent substrate for both serotonin and norepinephrine transporters, triggering neurotransmitter release, with comparatively weaker effects on dopamine (Rothman & Baumann, 2002). These mechanisms become particularly significant considering known sex differences in serotonergic function. Women show approximately 39% higher $5\text{-}HT_{1A}$ receptor binding throughout cortical and limbic regions while having 55% lower serotonin transporter binding compared to men (Jovanovic et al., 2008). These baseline differences could mean that when the system is challenged by psychedelic compounds, women experience more intense responses due to more readily available receptor sites, fewer transporters for clearance, and a system already optimized for heightened sensitivity to serotonergic modulation.

The substance-specific thematic divergences seen between *DAS* and *APA* align with MDMA and psilocybin's distinct mechanisms of action. Psilocybin's selective action on the serotonergic system, particularly 5-HT_{2A} receptors (Nichols, 2020), manifests in more frequent perceptual alterations and reality distortions. MDMA's broader impact on multiple neurotransmitter systems, including norepinephrine and dopamine (Freye, 2009; Verrico et al., 2006), key components of the body's "fight or flight" response, could contribute to more pronounced physiological symptoms. The higher rates of *bruxism* (37% versus 3%), *tremors* (37% versus 14%), and body temperature fluctuations (56% versus 24%) align with controlled studies showing increased adverse physiological effects in women (Liechti et al., 2001). Women's higher baseline parasympathetic tone and more rapid sympathetic activation to emotional stimuli (Verma et al., 2011) may explain the increased prevalence of autonomic symptom responses.

The widespread gastrointestinal distress reported by participants may also reflect sex-specific vulnerabilities as well. Women reported higher rates of *nausea* and *vomiting* for both psilocybin (45% and 34%) and MDMA (30% and 33%) compared to mixed-gender studies (12% nausea, Andersen et al., 2021; 24.2% nausea and 4.8% vomiting, Hinkle et al., 2024). This increased susceptibility may relate to menstrual cycle effects, as research indicates cyclic hormonal fluctuations can mediate susceptibility to nausea-inducing stimuli, with women not using oral contraception reporting higher nausea symptoms during the peri-menses phase (days 25-10) of their menstrual cycle (Matchock et al., 2008). This heightened sensitivity is potentially due to estrogen's influence on dopamine receptor activity, a system involved in nausea and vomiting mechanisms; notably, one with which both psilocybin and MDMA also interact (Nichols, 2004; Tylš et al., 2014).

The pervasive nature of *DI* underscores the importance of set and setting in shaping psychedelic experience (Hartogsohn, 2017). Most participants reported recreational intent and uncontrolled social environments which is in stark contrast to those typically provided in clinical trials (see Cavarra et al., 2022 for a systematic review). Despite 25% of participants recognizing the importance of mindset, few ensured the presence of a sober trip-sitter, and approximately one-third reported inadequate substance knowledge. These factors significantly contributed to adversity, particularly when substance effects diverged, failed to meet expectations or differed significantly from those of intoxicated peers. Notably, the familiarity of people or environment did not mitigate adverse experiences. Contextual variations and unpredictability frequently triggered *anxiety*, *fear*, and *paranoia*, especially when strangers were present. In some cases, familiar environments became hostile, leading to challenging *Interpersonal Dynamics* that sometimes prompted the desire for participants to flee or resist social support. These observations hold significant implications for clinical application, suggesting that standard protocols and therapeutic alliance may not always mitigate adversity of experience.

The nature of AAE observed in this study has been corroborated by a qualitative investigation of psilocybin-assisted therapy, where even within controlled environments, clients experience similar physiological, psychological, and perceptual disturbances (Nordin et al., 2024). Practitioners reported that clients may attempt to engage in ineffective coping strategies such as ruminative brooding or avoidance during difficult sessions, which, if not adequately addressed, can lead to ongoing cognitive and emotional instability. Notably, these AAE were also reported by participants as coping strategies, but were interpreted as an inability to self-regulate or losing control. This pattern is particularly significant for women, who show greater emotional reactivity, enhanced HPA axis activation, and are more likely to engage in ruminative thinking following stressful events (Verma et al., 2011; Goel et al., 2014).

Inexperience and psychedelic naivety have been associated with adverse experience in previous research (Carbonaro et al., 2016). The high proportion of participants reporting insufficient preparation, including nine first-time users, suggests that improved psychoeducation, preparation and integration support could mitigate some of the reported AAE. Several participants explicitly admitted to documenting their experience on Erowid to contribute knowledge and seek clarity about their own atypical reactions. The process of storytelling and seeking to (re)interpret a difficult experience is not uncommon amongst the psychedelic community and is often how users reframe "bad trips" into "challenging experiences" (Gashi et al., 2021). This narrative reframing serves as both a coping mechanism and an integration tool, enabling individuals to transform difficult experiences into sources of personal growth and incorporate them into their broader life stories (Carbonaro et al., 2016; Gashi et al., 2021). The role of supported integration therefore appears critical for optimizing therapeutic outcomes in clinical settings (Breeksema et al., 2022).

Methodologically, the LLM-assisted codebook review process revealed distinct patterns in coding agreement between the researcher and AI models. Claude demonstrated stronger alignment with researcher coding compared to ChatGPT across both substances. In terms of code frequency, Claude typically assigned more codes than ChatGPT but fewer than the researcher, while ChatGPT consistently applied the fewest codes. Report length appeared to significantly influence LLM code assignments - agreement levels varied substantially between reports, from 160 words (lowest agreement) to 1,302 words (highest agreement). While LLMs showed promise in supporting coding tasks, the observed discrepancies highlight the ongoing importance of human interpretation and validation in AI-assisted qualitative research. These findings align with recent studies exploring AI integration in qualitative methodologies (Dengel et al., 2023; Gao et al., 2023; Xiao et al., 2023) and suggest that while LLMs can be a valuable tool, it should be used to complement, rather than replace human analysis.

Several key limitations warrant consideration. The generalizability of findings is limited by non-probability sampling. The use of narratives from an online platform, independent of the study, means they are susceptible to voluntary reporting bias and risks of fabrication. The sample was relatively homogeneous, with ethnic diversity unascertainable and potential cultural bias due to the exclusive use of English across reports. The demographic skew towards younger participants may have influenced the prevalence of *OA*, as emotional regulation capabilities continue developing throughout young adulthood (Silvers et al., 2012). This developmental aspect is particularly relevant given younger individuals' increased vulnerability to challenging psychedelic experiences (Breeksema et al., 2022). Another significant limitation in this regard was the inability to examine hormonal influences, with menstrual cycle data provided by only one participant. This gap is particularly noteworthy as female risk of adverse drug reactions peaks between ages 30-39 and hormonal fluctuations are known to significantly impact drug response (Zucker & Prendergast, 2020).

The absence of control over substance identity, purity, and dosage was also a limitation, particularly relevant for MDMA experiences where adulteration is common in recreational contexts (Oliver et al., 2019; Pascoe et al., 2022). While results suggested comparable experiential adversity irrespective of substance type, the relatively modest sample size and potential lack of statistical power (.340) increase the risk of failing to detect real effects between substances. Future research should address these limitations through larger, controlled studies incorporating hormonal measurements, menstrual cycle tracking, and verified substance quality.

Psychedelic experiences are often characterized as highly idiosyncratic and context-dependent, a framing that raises valid concerns about their expanded clinical use (Aday et al., 2022; Andersen et al., 2020; Breeksema et al., 2022). While exploratory, this research has revealed a more consistent pattern of adverse female psychedelic experience that reinforces the critical need for more sex-specific considerations in psychedelic research. The divergences in thematic composition between psilocybin and MDMA indicate differentiated processes may be required for therapeutic protocols. As psychedelic-assisted therapy continues to evolve, these findings highlight the imperative for specialized therapist training in managing sex-specific adverse effects (Berger & Fitzgerald, 2023; Schlag et al., 2022). Attention to hormonal fluctuations, enhanced monitoring of physiological responses, and targeted integration practices that support emotional processing are recommended.

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Appendix A

Psychedelic Research Terms

Bad trip: a colloquial term used to describe an adverse drug taking experience, normally psychedelics. This is different to a *challenging experience* in that it is seen as a purely negative experience.

Challenging experience: describes a difficult or distressing drug taking experience that is differentiated from *bad trips* in that these experiences, while potentially frightening in the moment, are later integrated and reframed as valuable opportunities for insight, personal growth, and self-understanding.

Classic psychedelic: a psychoactive substance that primarily acts on serotonin 5-HT_{2A} receptors in the brain, producing profound alterations in perception, cognition, and consciousness. Classic psychedelics are characterized by their ability to induce non-ordinary states of consciousness, often described as mystical or transcendent experiences. Includes LSD, psilocybin, mescaline and DMT.

Derealization: a sense of feeling disconnected from the world around you.

Depersonalization: a sense of feeling disconnected from your own body, thoughts and feelings. Like you are watching what is happening to you as an outsider.

Entactogen: also known as an empathogen, is a class of psychoactive substances that produce experiences of emotional openness, empathy, and connectedness. These drugs primarily affect the release and reuptake of monoamine neurotransmitters (serotonin, dopamine, norepinephrine). Entactogens are distinct from classic psychedelics in their effects and mechanism of action.

Heroic dose: typically refers to a very large dose of a psychedelic substance; for psilocybin in the form of fungi, a heroic dose is often considered to be 5 grams or more of dried mushrooms. It is anticipated that a dose of this magnitude will produce a more intense psychedelic experience.

Integration: the ongoing process of sense making from and incorporating insights gained during psychedelic experiences into daily life. This can involve reflection, therapy, or personal practices to understand and apply these experiences in a beneficial and meaningful way (see Bathje et al., 2022 for a review and concept analysis).

Set: refers to the internal psychological factors that an individual brings to a drug experience. This includes their personality, mood, expectations, intentions, and mental preparation before using a substance. It encompasses both the person's immediate mindset, psychosocial education, experiences and longer-term attitudes and beliefs (Hartogsohn, 2017).

Setting: describes the external environmental factors surrounding a drug experience. This includes the physical space where the drug is taken, the social environment and interactions with others present, and the broader cultural context that shapes attitudes toward the substance. The setting influences how the drug effects are interpreted and experienced (Hartogsohn, 2017).

Trip: a common word used to describe an experience people have when taking a psychedelic or dissociative drug.

Trip sitter: a nominated, sober individual responsible for looking after someone who has taken a psychoactive or dissociative substance. Responsibilities may vary depending on the substance and nature of the trip but generally it is anticipated that they will provide general care and reduce the risk of a negative experience or other harms. This may include maintaining a sense of calm, monitoring and managing the environment to ensure it is safe and comfortable, providing emotional support, supervising activities, remaining alert to needs and providing water and food. Their role is not to provide psychedelic-assisted therapy.

Substances

Unless otherwise stated, definitions are from Alcohol and Drug Foundation (n.d.)

Ayahuasca: a powerful plant-based psychedelic drink that contains DMT (*dimethyltryptamine*) and MAOIs (monoamine oxidase inhibitors) as its active compounds. This concoction is primarily made from the *Banisteriopsis caapi* vine and *Psychotria viridis* leaves, though other plants may be incorporated. Profoundly affects perception, cognition, emotions, and time sense, often inducing hallucinations. Indigenous Amazonian peoples have used it for centuries in spiritual and medicinal contexts. Appears as a reddish-brown liquid with a strong, distinct odour and taste. It's known by various names across different cultures, including huasca, yagé, Kamarampi, Huni, brew, daime, the tea, and la purga.

Cannabis: a plant-derived substance containing various compounds called cannabinoids, primarily THC (*delta-9 tetrahydrocannabinol*) and CBD (*cannabidiol*). THC is the main psychoactive component that can alter mood and perception, while CBD is non-intoxicating. The *Cannabis sativa L* plant continues to be studied for its complex chemistry, with researchers still identifying new cannabinoids. While traditionally smoked, it can also be consumed through vaping, oils, or edibles. The same plant family includes hemp, which has minimal THC and is used industrially. Common street names include weed, pot, marijuana, hash, and dope.

DMT: (*N,N-Dimethyltryptamine*) a naturally-occurring psychedelic compound found in various plants and animals, as well as produced synthetically as a crystalline powder. Known for causing intense but relatively brief alterations in perception, consciousness, and emotional state. It is the active component in ayahuasca, a traditional brew that combines DMT-containing plants with others that allow for oral absorption. When used, DMT can produce significant changes in sensory experience, time perception, and thought patterns, often including vivid visual phenomena. Street names include Dimitri, The Spirit Molecule, and Changa.

LSD: (*lysergic acid diethylamide*) is a potent synthetic psychedelic drug derived from ergot, a fungus that grows on rye. It can induce significant changes in perception, mood, and cognition, with larger doses potentially causing visual hallucinations and distortions in the experience of space and time. Typically distributed on small squares of paper (tabs), sugar cubes, or gelatine sheets. It may also be found in liquid form or, less commonly, as tablets or capsules. Street names for LSD include acid, trips, tabs, microdots, dots, and Lucy.

MDMA: (*3,4-methylenedioxymethamphetamine*) a synthetic compound known as an empathogen due to its ability to increase feelings of emotional connection, empathy, and social bonding. Pure MDMA can appear as crystals, powder, or be pressed into tablets/capsules, though substances sold as "ecstasy" or "molly" may contain varying amounts of MDMA or other compounds. The variation in composition makes effects unpredictable. Street names include E, XTC, pills, and pingers.

Psilocybin: (*phosphoryloxy-N,N-dimethyltryptamine*) a naturally occurring psychedelic compound found in various species of mushrooms, commonly called "magic mushrooms." When consumed, psilocybin converts to psilocin in the body, producing significant changes in perception, thought patterns, emotional state, and sense of time. While traditionally consumed as fresh or dried mushrooms, it can also be synthesized as a white crystalline powder and prepared in capsules or tablets. Other names include shrooms, mushies, blue meanies, golden tops, liberty caps.

Appendix B

Table B1

Results of Coding Comparison between Researcher, Claude and ChatGPT (LLM)

		R	CL	GPT	R-CL	R-GPT	CL-GPT			
Erowid Report ID	Drug	Number of assigned codes			Number of matching codes			Pair-wise[a]	Total unique codes	Total code match[b]
93,948	P	7	13	7	4 (31%)	2 (29%)	2 (15%)	25%	20	1 (5%)
90,298	P	22	14	12	5 (23%)	7 (32%)	4 (29%)	28%	34	2 (6%)
95,189	P	32	41	11	16 (39%)	5 (16%)	8 (20%)	25%	59	4 (7%)
111,981	M	17	19	8	9 (47%)	3 (18%)	5 (26%)	30%	29	2 (7%)
5,303	M	9	13	8	4 (31%)	2 (22%)	2 (15%)	23%	24	2 (8%)
66,596	P	16	23	7	7 (30%)	5 (31%)	4 (17%)	26%	33	3 (9%)
55,285	P	21	19	8	11 (52%)	3 (14%)	8 (42%)	36%	29	3 (10%)
28,702	P	6	13	6	3 (23%)	2 (33%)	5 (38%)	32%	17	2 (12%)
50,799	P	42	25	8	17 (40%)	6 (14%)	8 (32%)	29%	50	6 (12%)
93,946	P	31	23	8	15 (48%)	5 (16%)	6 (26%)	30%	41	5 (12%)
41,948	M	10	15	7	4 (27%)	3 (30%)	4 (27%)	28%	24	3 (13%)
28,506	P	42	34	11	22 (52%)	8 (19%)	9 (26%)	33%	55	7 (13%)
933	M	16	15	8	11 (69%)	4 (25%)	4 (27%)	40%	24	4 (17%)
75,292	M	25	16	9	13 (52%)	7 (28%)	6 (38%)	39%	29	5 (17%)
72,786	P	14	14	10	7 (50%)	4 (29%)	8 (57%)	45%	23	4 (17%)
39,888	M	28	22	8	16 (57%)	7 (25%)	7 (32%)	38%	34	6 (18%)
20,417	M	11	16	9	7 (44%)	5 (45%)	6 (38%)	42%	22	4 (18%)
758	M	17	16	7	10 (59%)	5 (29%)	7 (44%)	44%	23	5 (22%)
3,214	M	29	18	10	14 (48%)	9 (31%)	9 (50%)	43%	34	9 (26%)
4,207	M	9	10	7	9 (90%)	7 (78%)	7 (70%)	79%	10	7 (70%)
Average P		23	22	9	11 (39%)	5 (23%)	6 (30%)	33%	36	4 (13%)

Average M	17	16	8	10 (52%)	5 (33%)	6 (37%)	33%	25	5 (13%)
Average total	20	19	8	10 (46%)	5 (28%)	6 (33%)	36%	31	4 (16%)

Note. M = MDMA; P = psilocybin; R = researcher; CL = Claude 3.5 Sonnet; GPT = ChatGPT-4.

[a] Average pairwise agreement i.e. the average exact code agreement between each pair of coders.

[b] Overall inter-coder agreement i.e. percentage of exact code agreement across all three coders.

Table B2

A Comparison of Final Codes Assigned by Researcher, Claude and ChatGPT to Representative Reports

Erowid Report ID	Agreement level	Researcher	Claude (3.5 Sonnet)	ChatGPT
93,948	Lowest (5%)	Disembodiment and derealization; Disgust; Gastrointestinal distress; Guilt; Hallucinations; Paranoia; Self-consciousness.	Abrupt onset of challenging effects; Afraid of social repercussions; Gastrointestinal distress; Guilt; Impaired mobility; Increased exteroception; Loss of appetite; Paranoia; Self-consciousness; Somatic-centric experience; Subterfuge employed; Uncomfortable; Unease.	Bruxism and teeth chattering; Cold body temperature; Fear; Gastrointestinal distress; Hallucinations; Somatic-centric experience; Tense muscles.
758	Moderate (22%)	Abrupt onset of challenging effects; Bruxism and teeth chattering; Dilated pupils; Disembodiment and derealization; Emotional whiplash or rollercoaster; Fear; Frustration; Gastrointestinal distress; Hallucinations; Loss of appetite; Loss of control over body; Overwhelming discomfort; Overwhelming physiological response;	Abrupt onset of challenging effects; Bruxism and teeth chattering; Dilated pupils; Fatigue; Gastrointestinal distress; Hallucinations; Impulsive redosing; Increased exteroception; Interrupted sleep; Loss of appetite; Overwhelming physiological response; Response to drug incongruent to expectations; Shaking,	Bruxism and teeth chattering; Dilated pupils; Fatigue; Gastrointestinal distress; Hallucinations; Shaking body spasms and seizures; Vision impairment.

		Shaking, body spasms and seizures; Somatic-centric experience; Sweating; Temporal distortions.	body spasms and seizures; Somatic-centric experience; Temporal distortions; Vision impairment.	
4,207	Highest (70%)	Bruxism and teeth chattering; Cold (body temperature); Fear; Gastrointestinal distress; Hallucinations; Loss of control over body; Overwhelming physiological response; Somatic-centric experience; Tense muscles	Abrupt onset of challenging effects; Bruxism and teeth chattering; Cold (body temperature); Fear; Gastrointestinal distress; Hallucinations; Loss of control over body; Overwhelming physiological response; Somatic-centric experience; Tense muscles	Bruxism and teeth chattering; Cold (body temperature); Fear; Gastrointestinal distress; Hallucinations; Somatic-centric experience; Tense muscles

Note. Codes are drawn from the researcher generated codebook; LLM's were not used to generate codes for the coding agreement comparison.

Table B3

Code Agreement for Erowid Reports Between Researcher, Claude and ChatGPT in Order of Total Exact Code Agreement from Lowest to Highest

Erowid Report ID	Codes
93,948	Gastrointestinal distress
90,298	Disembodiment and derealization; Fear
95,189	Hallucinations; Panic; Racing thoughts; Sense of evil presence
111,981	Hallucinations; Shaking, body spasms and seizures
5,303	Hallucinations; Vision impairment
66,596	Fear; Gastrointestinal distress; Hallucinations
55,285	Crying; Gastrointestinal distress; Hallucinations
28,702	Fear; Racing thoughts
50,799	Despair; Fear; Loss of control; Panic; Sense of being possessed or taken over; Suicidal ideation

93,946	Anxiety; Gastrointestinal distress; Hallucinations; Overwhelming discomfort; Racing thoughts
41,948	Gastrointestinal distress; Shaking, body spasms and seizures; Sweating
28,506	Abrupt onset of challenging effects; Confusion; Gastrointestinal distress; Hallucinations; Overwhelming discomfort; Panic; Racing thoughts
933	Fear; Hot (body temperature); Shaking, body spasms and seizures; Sweating
75,292	Chest pains; Fear; Hot (body temperature); Panic; Sweating
72,786	Confusion; Fear; Gastrointestinal distress; Hallucinations
39,888	Abrupt onset of challenging effects; Cold (body temperature); Fear; Gastrointestinal distress; Hallucinations; Shaking, body spasms and seizures
20,417	Emotional whiplash or rollercoaster; Fear; Overwhelmed; Sweating
758	Bruxism and teeth chattering; Dilated pupils; Gastrointestinal distress; Hallucinations; Shaking, body spasms and seizures
3,214	Cold (body temperature); Confusion; Disembodiment and derealization; Emotional whiplash or rollercoaster; Fear; Hallucinations; Lost; Overwhelmed; Vision impairment
4,207	Bruxism and teeth chattering; Cold (body temperature); Fear; Gastrointestinal distress; Hallucinations; Somatic-centric experience; Tense muscles

Appendix C

Demographic Characteristics of Total Participant Population and Individual Substance Subgroups of MDMA and Psilocybin

Characteristic	Total (*N* = 56)	MDMA (*n = 27*)	Psilocybin (*n* = 29)
Age (in years)			
15 - 19	8 (14.3%)	3 (11.1%)	5 (17.2%)
20 - 24	4 (7.1%)	2 (7.4%)	2 (6.9%)
25 - 29	1 (1.8%)	-	1 (3.4%)
30 - 34	-	-	-
35 - 39	-	-	-
40 - 45	1 (1.8%)	-	1 (3.4%)
Not disclosed	42 (75%)	22 (81.5%)	20 (69%)
Mean age	19 (*SD* = 7.3)	20 (*SD* = 2.0)	18 (*SD* = 9.1)
Weight (in kgs)			
40 – 49	14 (25%)	6 (22.2%)	8 (27.6%)
50 – 59	26 (46.4%)	13 (48.1%)	13 (44.8%)
60 - 69	8 (14.3%)	4 (14.8%)	13 (44.8%)
70 - 79	3 (5.4%)	1 (3.7%)	2 (6.9%)
80 - 89	-	-	-
100 - 109	-	-	-
110 - 119	1 (1.8%)	1 (3.7%)	0
Not disclosed	4 (7.1%)	2 (7.4%)	2 (6.9%)
Mean weight	56 (*SD* = 11.8)	57 (*SD* = 14.5)	56 (*SD* = 8.9)
Relationship status			
Single	2 (3.6%)	-	2 (6.9%)
Boyfriend	17 (30.4%)	9 (33.3%)	8 (27.6%)
Husband	2 (3.6%)	1 (3.7%)	1 (3.4%)

Separated	1 (1.8%)	-	1 (3.4%)
Not disclosed	34 (60.7%)	17 (63%)	17 (58.6%)
Primary employment status			
In paid employment	6 (10.7%)	1 (3.7%)	5 (17.2%)
Student	10 (17.9%)	5 (18.5%)	5 (17.2%)
Unemployed	1 (1.8%)	-	1 (3.4%)
Not disclosed	39 (69.6%)	21 (77.8%)	18 (62.1%)
Pre-existing mental health condition			
Depression	5 (8.9%)	1 (3.7%)	4 (13.8%)
Not disclosed	51 (91.1%)	26 (96.3%)	25 (86.2%)

Appendix D

Dose, Previous Substance Experience, Set and Setting Characteristics of Total Participant Population and Individual Substance Subgroups of MDMA and Psilocybin

Characteristic	Total (*N* = 56)	MDMA (*n = 27*)	Psilocybin (*n* = 29)
Dose			
Common: 1g – 2.5g	4 (7.1%)	N/A	4 (7.1%)
Strong: 2.5g – 5g	14 (25%)	N/A	14 (25%)
Heavy: 5g +	6 (10.7%)	N/A	6 (10.7%)
0.5 tablet	3 (5.4%)	N/A	
1 tablet	12 (21.4%)	12 (21.4%)	N/A
1.5 tablet	1 (1.8%	1 (1.8%	N/A
2 tablets	4 (7.1%	4 (7.1%	N/A
3 tablets	2 (2.6%)	2 (2.6%)	N/A
Indeterminate	10 (17.8%)	5 (8.9%)	5 (8.9%)
Consumption mode			
Oral	55 (98.2%)	26 (96.3%)	29 (100%)
Insufflated	1 (1.8%)	1 (1.8%)	-
Drug experience			
First time with RD	15 (26.8%)	7 (25.9%)	8 (27.6%)
Pre-trip experience with RD	32 (57.1%)	17 (63%)	15 (51.7%)
Pre-trip positive experience with RD	12 (21.4%)	3 (11.1%)	9 (31%)
Pre-trip drug use experience (general)	45 (80.4%)	20 (74.1%)	25 (86.2%)
Pre-trip psychedelic experience	35 (62.5%)	16 (59.3%)	19 (65.5%)
Pre-trip cannabis experience	14 (25%)	3 (11.1%)	11 (37.9%)
Mindset (trip intention)			

Escapism	1 (1.8%)	-	1 (1.8%)
Personal exploration	1 (1.8%)	-	1 (1.8%)
Recreational	36 (64.3%)	25 (92.6%)	11 (37.9%)
Recreational and experimental	10 (17%)	2 (7.4%)	8 (27.6%)
Recreational and personal exploration	6 (10.7%)	-	6 (20.7%)
Spiritual	1 (1.8%)	-	1 (1.8%)
Spiritual and therapeutic	1 (1.8%)	-	1 (1.8%)
Awareness of set importance			
Acknowledged	14 (25%)	5 (18.5%)	9 (31%)
No indication given	42 (75%)	22 (81.5%)	(20 (69%)
Indications of inadequate preparation	20 (35.7%)	7 (25.9%)	13 (44.8%)
Lazare Faire attitude to dosage	11 (19.6%)	5 (18.5%)	6 (20.7%)
Trip-sitter present	4 (7.1%)	1 (3.7%)	3 (10.3%)
Drug taking cohort			
Alone	1 (1.8%)	-	1 (3.4%)
Partner	8 (14.3%)	2 (7.4%)	6 (20.7%)
Friend(s)	28 (50%)	15 (55.6%)	13 (44.8%)
Family (sibling)	2 (3.6%)	1 (3.7%)	1 (3.4%)
Friend(s) and partner	15 (26.8%)	8 (29.6%)	7 (24.1%)
Friend(s) and family (sibling)	1 (1.8%)	-	1 (3.4%)
No indication given	1 (1.8%)	1 (3.7%)	-
Social context			
Solo	1 (1.8%)	-	1 (3.4%)
2 people	14 (25%)	5 (18.5%)	9 (31%)
3 people	4 (7.1%	-	4 (13.8%)
Variable	36 (64.3%)	21 (77.8%)	15 (51.7%)
Small group (4 – 6)	10 (17.9%)	3 (11.1%)	3 (10.3%)

Medium group (7 – 15)	6 (10.7%)	3 (11.1%)	3 (10.3%)
Large group (16 – 30)	1 (1.8%)	1 (3.7%)	-
Gathering (30 – 50)	2 (3.6%)	1 (3.7%)	1 (3.4%)
Crowd (50 – 300)	1 (1.8%)	1 (3.7%)	-
Mass (300 – 1000+)	5 (8.9%)	5 (18.5%)	-
No indication given	1 (1.8%	1 (3.7%)	-
Country of experience			
Canada	1 (1.8%)	-	1 (3.4%)
Netherlands	1 (1.8%)	-	1 (3.4%)
Scotland	1 (1.8%)	-	1 (3.4%)
United States of America	2 (3.6%	-	2 (6.9%
No indication given	51 (91.1%)	27 (100%)	24 (82.8%)
Location of most AAE			
Across all environments	3 (5.4%)	1 (3.7%)	2 (6.9%)
Home environments	32 (57.1%)	10 (37%)	22 (75.9%)
Public spaces	15 (26.8%)	12 (44.4%)	3 (10.3%)
Outdoors in nature	1 (1.8%)	0	1 (3.4%)
In moving vehicle	4 (7.1%)	3 (11.1%)	1 (3.4%)
No indication given	1 (1.8%)	1 (3.7%)	-
Menstruating	1 (1.8%)	-	1 (3.4%)
Medical attention			
Ambulance or medic	3 (5.4%)	1 (3.7%)	2 (6.9%)
A&E	2 (3.6%)	1 (3.7%)	1 (3.4%)
Hospitalisation	1 (1.8%)	1 (3.7%)	-
No indication given	50 (89.3%)	24 (88.9%)	26 (89.7%)

Note. RD = report drug; N/A = Not applicable; AAE = acute adverse effects; to remain consistent with participant accounts MDMA dose is reported in tablet numbers and psilocybin dose is reported in grams; all psilocybin was consumed in the form of fungi. In this form psilocybin and psilocin potency can vary significantly both across and within mushroom species. *Psilocybe cubensis*, a common species for non-clinical use, typically contains 10–12 mg of psilocybin per gram of dried mushrooms (Psilocybin, 2016); *Drug experience* is based on participant disclosure therefore pre-existing drug experience may be underrepresented; *Pre-trip drug use experience (general)* includes indications of previously consuming one or more of the following drugs: MDMA, psilocybin, benzodiazepines, cannabis, cocaine, heroin, LSD or methamphetamine. Based on the nature of the data, geographical locations were understood as experience location rather than country of residence; *Home environments* includes both personal or a friend's home; *Public spaces* includes a café, movie theatre, bars, clubs, raves or parties. *Menstruating* refers to participants who indicated they were menstruating at the time of the experience.

Appendix E

Prevalence of Specific Acute Adverse Effects Reported for MDMA and Psilocybin

AAE	Total (*N* = 56)	MDMA (*n = 27*)	Psilocybin (*n* = 29)
Abdominal discomfort	10 (18%)	4 (17%)	6 (21%)
Abnormal bodily sensations	42 (75%)	20 (74%)	22 (76%)
Abnormal taste in mouth	2 (4%)	-	2 (7%)
Agitation or restlessness	13 (23%)	7 (26%)	6 (21%)
Aloneness	12 (21%)	3 (11%)	9 (31%)
Anxiety	33 (59%)	16 (56%)	17 (59%)
Appetite change	1 (2%)	1 (4%)	-
Blacking out/fainting	6 (11%)	3 (11%)	3 (10%)
Blurred or unfocused vision	9 (16%)	5 (19%)	4 (14%)
Bruxism	11 (20%)	10 (37%)	1 (3%)
Change in speed of thoughts	14 (25%)	4 (15%)	10 (34%)
Chest pain or pressure	3 (5%)	3 (11%)	-
Concentration or memory difficulty	11 (20%)	2 (7%)	9 (31%)
Crying and tearfulness	21 (38%)	6 (22%)	15 (52%)
Dehydration and salivation changes	5 (9%)	2 (7%)	3 (10%)
Depersonalization	15 (27%)	2 (7%)	13 (45%)
Derealization, dissociation, or sense of surrealness	32 (57%)	10 (37%)	22 (76%)
Diarrhoea	-	-	-
Difficulty sleeping	14 (25%)	7 (26%)	7 (24%)
Disorientation or confusion	26 (46%)	7 (26%)	19 (66%)
Dyspnea	12 (21%)	6 (22%)	6 (21%)
Dull feeling	1 (2%)	1 (4%)	-
Fatigue or weakness	13 (23%)	7 (26%)	6 (21%)
Fear, distress or despair	55 (98%)	26 (96%)	29 (100%)
Feeling cold	13 (23%)	8 (30%)	5 (17%)

Feeling hot	9 (16%)	7 (26%)	2 (7%)
Feeling of wetness	3 (5%)	1 (4%)	2 (7%)
Hallucination	28 (50%)	8 (30%)	20 (69%)
Headache or migraine	3 (5%)	3 (11%)	-
Hearing change or tinnitus	6 (11%)	2 (7%)	11 (38%)
Helplessness	19 (34%)	7 (26%)	12 (41%)
Hopelessness	9 (16%)	2 (7%)	7 (24%)
Hyperactivity	11 (20%)	3 (11%)	8 (28%)
Hypertension	-	-	-
Illusion	24 (43%)	4 (15%)	20 (69%)
Impaired speech abilities	13 (23%)	2 (7%)	11 (38%)
Incoordination	16 (29%)	7 (26%)	9 (31%)
Laughter	4 (7%)	1 (4%)	3 (10%)
Lightheadedness, dizziness and vertigo	6 (11%)	5 (19%)	1 (3%)
Mood depression	5 (9%)	3 (11%)	2 (7%)
Mood lability or change	15 (27%)	5 (19%)	10 (34%)
Musculoskeletal discomfort	7 (13%)	4 (15%)	3 (10%)
Mydriasis (dilated pupils)	5 (9%)	2 (7%)	3 (10%)
Nausea	21 (38%)	8 (30%)	13 (45%)
Odour sensitivity	-	-	-
Oscillating body temperature	3 (5%)	2 (7%)	1 (3%)
Palpitations	-	-	-
Panic	36 (64%)	13 (48%)	23 (79%)
Paranoia	23 (41%)	9 (33%)	14 (48%)
Paresthesias or numbness	12 (21%)	5 (19%)	7 (24%)
Perspiration	16 (29%)	10 (37%)	6 (21%)
Seizures	4 (7%)	3 (11%)	1 (3%)
Skin colour changes	4 (7%)	3 (11%)	1 (3%)

Somnolence	8 (14%)	4 (15%)	4 (14%)
Suicidal ideation	2 (4%)	-	2 (7%)
Synesthesia	1 (2%)	-	1 (3%)
Tachycardia	8 (14%)	4 (15%)	4 (14%)
Temporal distortions	21 (38%)	5 (19%)	16 (55%)
Tremulousness	14 (25%)	10 (37%)	4 (14%)
Urinary symptoms	7 (13%)	4 (15%)	3 (10%)
Vomiting	19 (34%)	9 (33%)	10 (34%)
Yawning	4 (7%)	-	4 (14%)

Note. AAE = acute adverse effects; frequencies denote the number of individual reports where a specific AAE was present i.e. how many individual participants experienced the AAE, not how often it occurred across reports.